



Matrix Theory

13 Orthogonal Complement

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Fundamental theorem of Orthogonality



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The row space is orthogonal to null space (in $\mathbb{R}^n$). The column space is orthogonal to the left null space (in $\mathbb{R}^m$).

Example



$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}_{3 \times 2}$$

$\mathbb{R}^2$

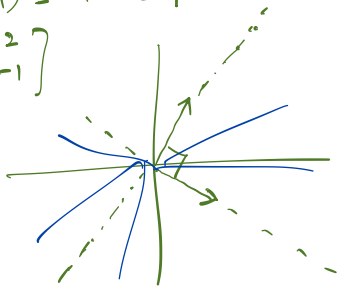


$\mathbb{R}^3$

$$C(A^T) = C\left[\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right] \quad r=1$$

$$N(A) = n - r = 1$$

$$K\left[\begin{pmatrix} 2 \\ -1 \end{pmatrix}\right]$$



Observations



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- $N(A) \perp C(A^T)$

Observations



- $N(A) \perp C(A^T)$ ✓
- In fact, $N(A)$ contains every vector perpendicular to $C(A^T)$

$$N(A) = \{x: Ax = 0\}$$



- In general, the dimension of $N(A)$ and $C(A^T)$ add to the dimension of the ambient space

Observations



$$\eta - r + r = \eta$$

- In general, the dimension of $N(A)$ and $C(A^T)$ add to the dimension of the ambient space
- Similarly for $N(A^T)$ and $C(A)$

$$\text{if } \underline{a \perp N(A)} \quad \underline{\eta - r}$$

$$a \in C(A^T) \quad \underline{r}$$

Orthogonal Complement



Given a subspace $\mathbf{V}$ of $\mathbb{R}^n$, the space of all vectors orthogonal to $\mathbf{V}$ is called the Orthogonal Complement of $\mathbf{V}$. It is denoted by $\mathbf{V}^\perp = \mathbf{V}$ perp.

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- $N(A) = (C(A^T))^\perp$

Orthogonal Complement



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- Also, the row space contains every vector orthogonal to null space,
 $C(A^T) = (N(A))^\perp$

Orthogonal Complement



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- $N(A) = (C(A^T))^\perp$
- Also, the row space contains every vector orthogonal to null space,
 $C(A^T) = (N(A))^\perp$
- Similarly for the $N(A^T)$ and $C(A)$ pair.

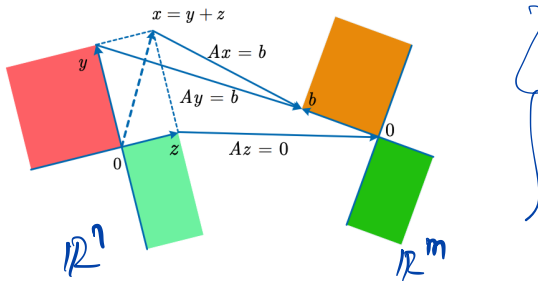
Fundamental Theorem of LA, Part II



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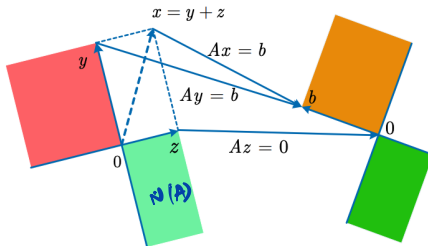
The nullspace is the orthogonal complement of the row space in $\mathbb{R}^n$.
The left nullspace is the orthogonal complement of the column space in $\mathbb{R}^m$.

Some notes



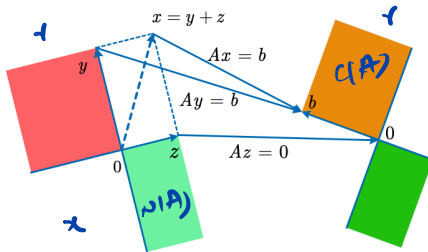
- The matrix can operate on any input in the ambient space

Some notes



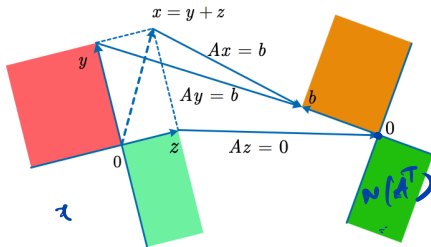
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- Null space is carried to zero

Some notes



- The matrix can operate on any input in the ambient space
- Null space is carried to zero
- Every vector Ax is in the column space

Some notes



$$A^T y = 0 \quad \checkmark$$

$$\underline{Ax = y} \quad \checkmark$$

- The matrix can operate on any input in the ambient space
- Null space is carried to zero
- Every vector Ax is in the column space
- Nothing goes to $N(A^T)$

$$A^T Ax = 0$$

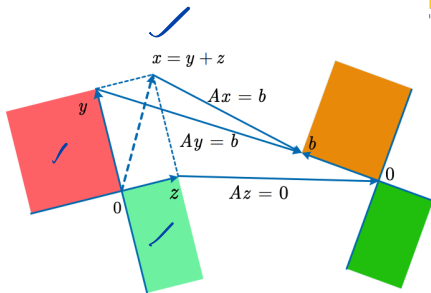
$$x^T A^T Ax = 0$$

$$\|Ax\|^2 = 0$$

$$Ax = 0$$

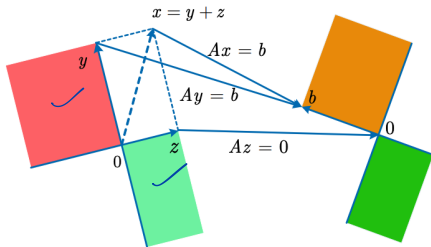


Some notes



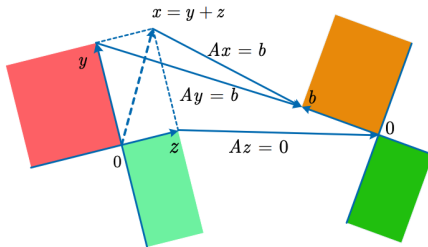
- There can be vectors outside $C(A^T)$ and $N(A)$ (i.e., their union need not fill all of $\mathbb{R}^n$).

Some notes



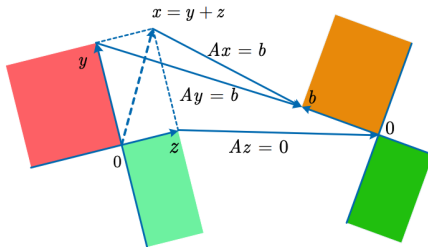
- There can be vectors outside $C(A^T)$ and $N(A)$ (i.e., their union need not fill all of $\mathbb{R}^n$).
- However, their sum does ✓

Some notes



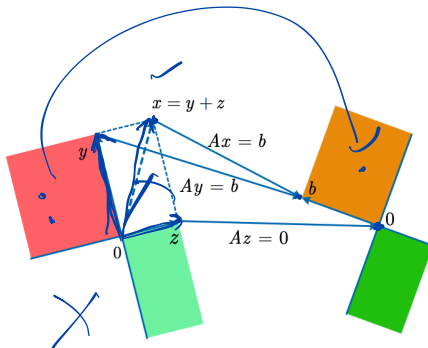
- Every x splits uniquely as $x = \underline{x_r} + \underline{x_n}$, with x_r in the row space and x_n in the nullspace

Some notes



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- x_n goes to zero: $Ax_n = 0$

Some notes



- Every x splits uniquely as $x = x_r + x_n$, with x_r in the row space and x_n in the nullspace
- x_n goes to zero: $Ax_n = 0$
- x_r goes to $C(A)$: $Ax_r = Ax$

Some notes



From the $C(A^T)$ to $C(A)$, matrix A is invertible. Every vector b in the column space comes from exactly one vector x_r in the row space.

$$A x_r = b$$

$$A x'_r = b$$

$$A (x_r - x'_r) = 0$$

↓



- Projections

Rough Work



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